

# Prajwal Khanal

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| RESEARCH INTERESTS | Remote sensing of vegetation and its water use; drought; hybrid process-based and machine-learning models.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                |
| EDUCATION          | <b>PhD, Geo-Information Science and Earth Observation, University of Twente</b><br>– WUNDER: field-scale soil moisture and fluxes for agricultural drought monitoring.<br>– Supervisors: Prof. dr. Bob Su and dr. Yijian Zeng.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Feb 2023—...                   |
|                    | <b>M.Sc., Environmental Engineering, Technical University of Munich</b><br>– Graduated 1.4/5.0 (1 = best). Thesis: soil moisture controls on vegetation, in <i>Biogeosciences</i> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Nov 2020—Dec 2022              |
|                    | <b>B.E., Civil Engineering, Institute of Engineering, Tribhuvan University, Nepal</b><br>– Ranked 20/192, 77.7%. Thesis: flood inundation mapping, Babai River.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Nov 2014—Aug 2018              |
| EMPLOYMENT         | <b>Research Intern, Max Planck Institute for Biogeochemistry, Jena</b><br>– Near-surface soil moisture vs. terrestrial water storage, controls on global vegetation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Nov 2020—Feb 2022              |
|                    | <b>Student Research Assistant, Technical University of Munich</b><br>– Flood inundation maps for the Ouémé (Benin) and Naryn (Kyrgyzstan) rivers, HEC-RAS.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Nov 2021—Mar 2022              |
|                    | <b>Working Student, Jena-Geos Ingenieurbüro GmbH, Munich</b><br>– Excel/VBA tool for greenhouse-gas accounting: combustion, refrigeration, transport.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | May 2021—Dec 2021              |
|                    | <b>Civil Engineer (freelance), ECEP, Kathmandu, Nepal</b><br>– Business plans for community water supply projects, Government of Nepal.<br>– Smart water supply master plan for Lumbini, Ministry of Urban Development.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Dec 2018—Dec 2022              |
| PUBLICATIONS       | <i>Peer-reviewed</i><br>Moutahir, H., Sulzer, M., Kiese, R., ... <b>Khanal, P.</b> , et al. (2026). <u>Matching scales of eddy covariance measurements and process-based modelling: assessing spatiotemporal dynamics of carbon and water fluxes in a mixed forest in Southern Germany</u> . <i>Biogeosciences</i> , 23, 1719–1738.<br>Zeng, Y., Alidoost, F., Schilperoort, B., ... <b>Khanal, P.</b> , et al. (2025). <u>Towards an open soil-plant digital twin based on the STEMMUS-SCOPE model, following open science</u> . <i>Computers &amp; Geosciences</i> , 205, 106013.<br><b>Khanal, P.</b> , Hoek van Dijke, A. J., Schaffhauser, T., et al. (2024). <u>Relevance of near-surface soil moisture vs. terrestrial water storage for global vegetation functioning</u> . <i>Biogeosciences</i> , 21(6), 1533–1547.<br>Bhattarai, P., <b>Khanal, P.</b> , Tiwari, P., et al. (2019). <u>Flood inundation mapping of Babai basin using HEC-RAS and GIS</u> . <i>Journal of the Institute of Engineering</i> , 15(2), 32–44.<br><i>Under review</i><br><b>Khanal, P.</b> , Zeng, Y., Daoud, M. G., Beckers, J., Su, Z. <u>Investigating the impact of multilayer resistance schemes on simulating land surface temperature and turbulent fluxes: evaluation within the STEMMUS-SCOPE model</u> . Preprint on SSRN.<br><b>Khanal, P.</b> , et al. <u>Deep learning soil moisture and fluxes with multimodal foundation-model embeddings</u> .<br><i>Conference contributions</i><br><b>Khanal, P.</b> , et al. (2026). <u>Bridging the scale gap: daily 10-metre soil moisture, evapotranspiration and carbon fluxes via multimodal deep learning</u> . <i>Workshop on Machine Learning for Land and Hydrology</i> , ECMWF, Reading, UK.<br><b>Khanal, P.</b> , Hoek van Dijke, A., Zeng, Y., Orth, R. (2023). <u>Near-surface vs. sub-surface soil moisture impacts on vegetation functioning</u> . <i>EGU General Assembly</i> , Vienna, EGU23-7670.<br><b>Khanal, P.</b> (2019). <u>Flood inundation mapping: an inference on the outbreak of flood-borne diseases</u> . <i>International Conference on Health Engineering in Disaster</i> . |                                |
| AWARDS             | <b>Second place, Hackathon on Machine Learning for Earth System Modelling</b><br>– Quantization in foundation models. CESOC, ECMWF, TRA Modelling, FZ Jülich, Cologne.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Aug 2026                       |
| SCHOLARSHIPS       | <b>Deutschlandstipendium, merit scholarship, Technical University of Munich</b><br><b>Merit-based scholarship, Institute of Engineering, Tribhuvan University</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2021/22<br>2014–2018           |
| TEACHING           | <b>Teaching Assistant, Technical University of Munich</b><br>– River Basin Modelling; Integrated Water Resources Management; Rapidly Varying Flow.<br><b>Assistant Lecturer, Cosmos College of Management and Technology, Nepal</b><br>– Elective: Python and GIS for water resources.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2021–2022<br>Nov 2018—Jan 2021 |
| SOFTWARE           | <b>WUNDER data viewer, ITC, University of Twente</b><br>– Weather and soil moisture loggers, read by growers at the food forests as drought develops. <a href="#">Source</a> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2026                           |
| SKILLS             | <i>Modelling:</i> STEMMUS-SCOPE, land surface models, HEC-RAS, SWAT<br><i>Programming:</i> Python, PyTorch, R, GIS, Excel/VBA<br><i>Languages:</i> English (C2), German (A1), Dutch (A2), Hindi, Nepali                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                |